SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

Course Code: CSA07 **Course Title:** Computer Networks

CO Assessed: CO5 – K3 – Articulation **Assessment:** Assessment Tool 2

Student Name: ________________________________

Reg. No.: _________________________________

Section / Batch: ____________________________

Date: ______________________________________

Submission format: Typed A4 document containing the arranged pseudocode and the corresponding flowchart for Q1, Q2 and Q3.

Teacher instruction: Submit the GitHub folder/repository link in Google Classroom (GCR).

QUESTION 1: P2P FILE SHARING ALGORITHM

Topic: P2P Communication

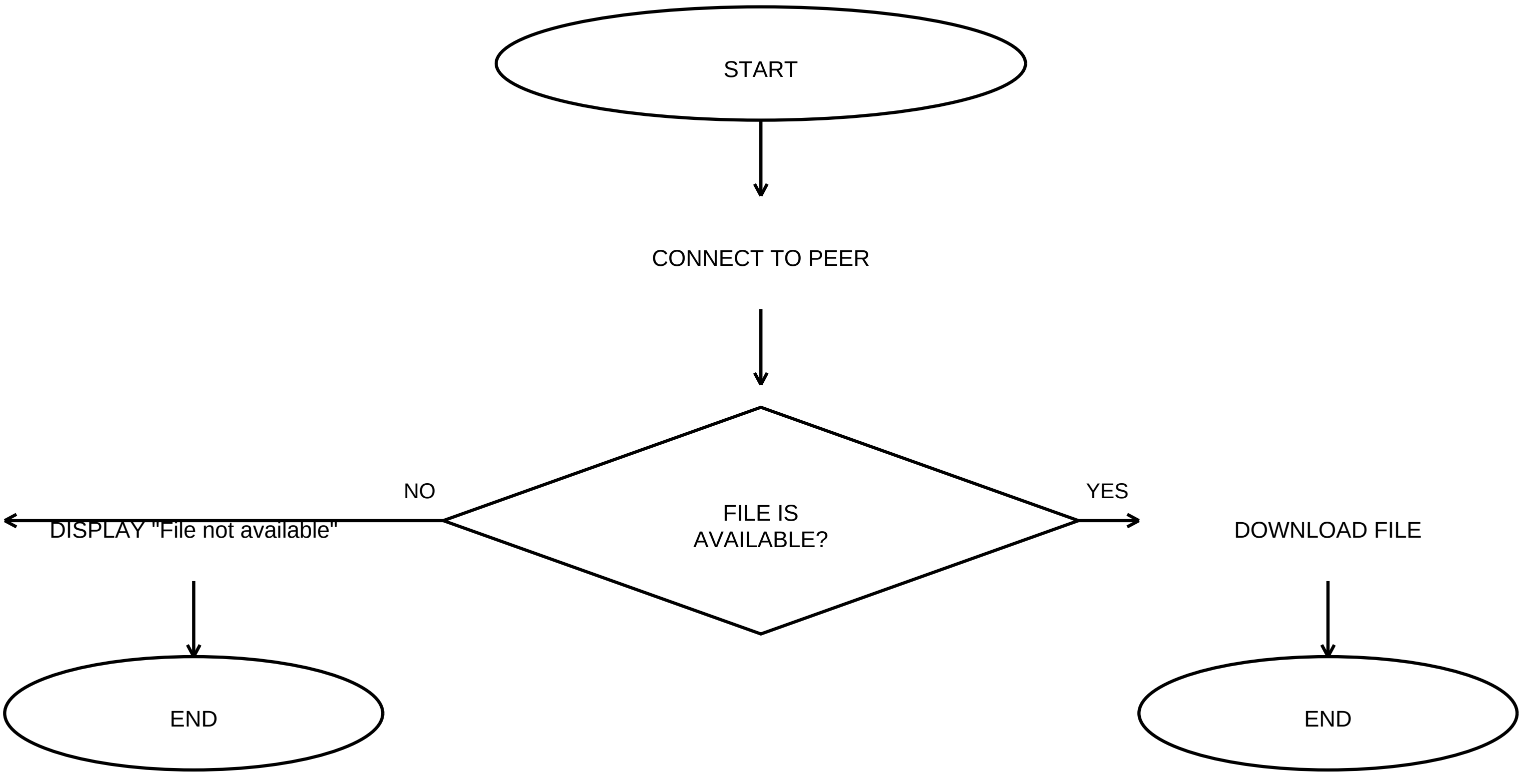
Scenario: A student wants to download a file from another computer in a peer-to-peer network.

A. Correct Pseudocode Block Arrangement

1. `START`
2. `CONNECT TO PEER`
3. `IF FILE IS AVAILABLE THEN`
4. `DOWNLOAD FILE`
5. `ENDIF`
6. `DISPLAY "File not available"`
7. `END`

Note about Q1: *The supplied block list has no ELSE block. Therefore, it cannot express the unavailable-file case correctly while also using every listed block unchanged. The flowchart shows the intended logic from the scenario: YES → download; NO → display the failure message. If the teacher requires the blocks to be used literally exactly once, ask whether an ELSE block was accidentally omitted.*

B. Corresponding Flowchart


START
CONNECT TO PEER
FILE IS AVAILABLE?
NO
YES
DISPLAY "File not available"
DOWNLOAD FILE
END
END

QUESTION 2: VoIP CALL CONNECTION ALGORITHM

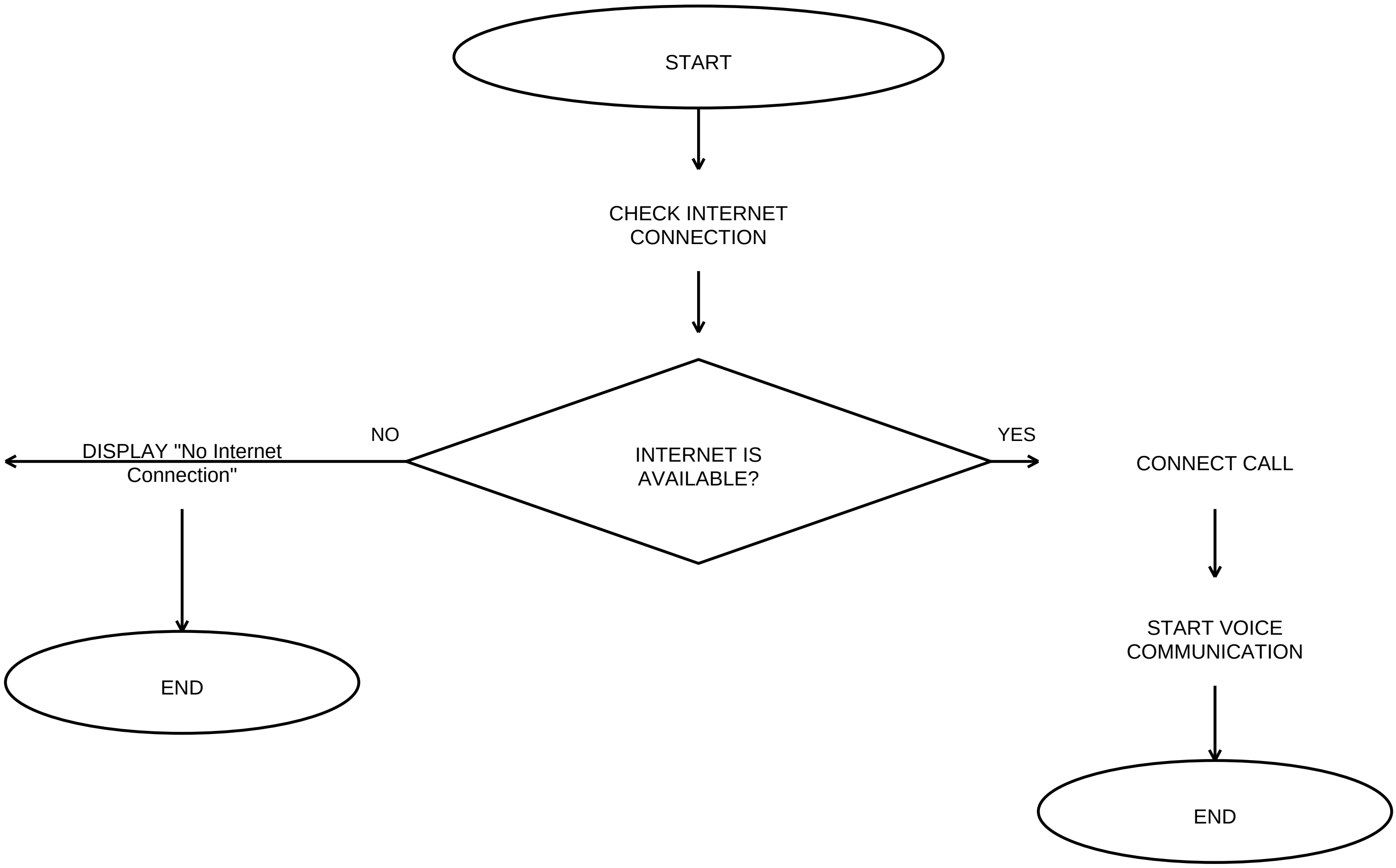
Topic: VoIP Communication

Scenario: A user starts a voice call using a VoIP application. The application checks the Internet connection before starting the call.

A. Correct Pseudocode Block Arrangement

1. `START`
2. `CHECK INTERNET CONNECTION`
3. `IF INTERNET IS AVAILABLE THEN`
4. `CONNECT CALL`
5. `START VOICE COMMUNICATION`
6. `ELSE`
7. `DISPLAY "No Internet Connection"`
8. `ENDIF`
9. `END`

B. Corresponding Flowchart


START
CHECK INTERNET CONNECTION
INTERNET IS AVAILABLE?
NO
YES
DISPLAY "No Internet Connection"
END
CONNECT CALL
START VOICE COMMUNICATION
END

QUESTION 3: OVERLAY NETWORK MESSAGE ROUTING ALGORITHM

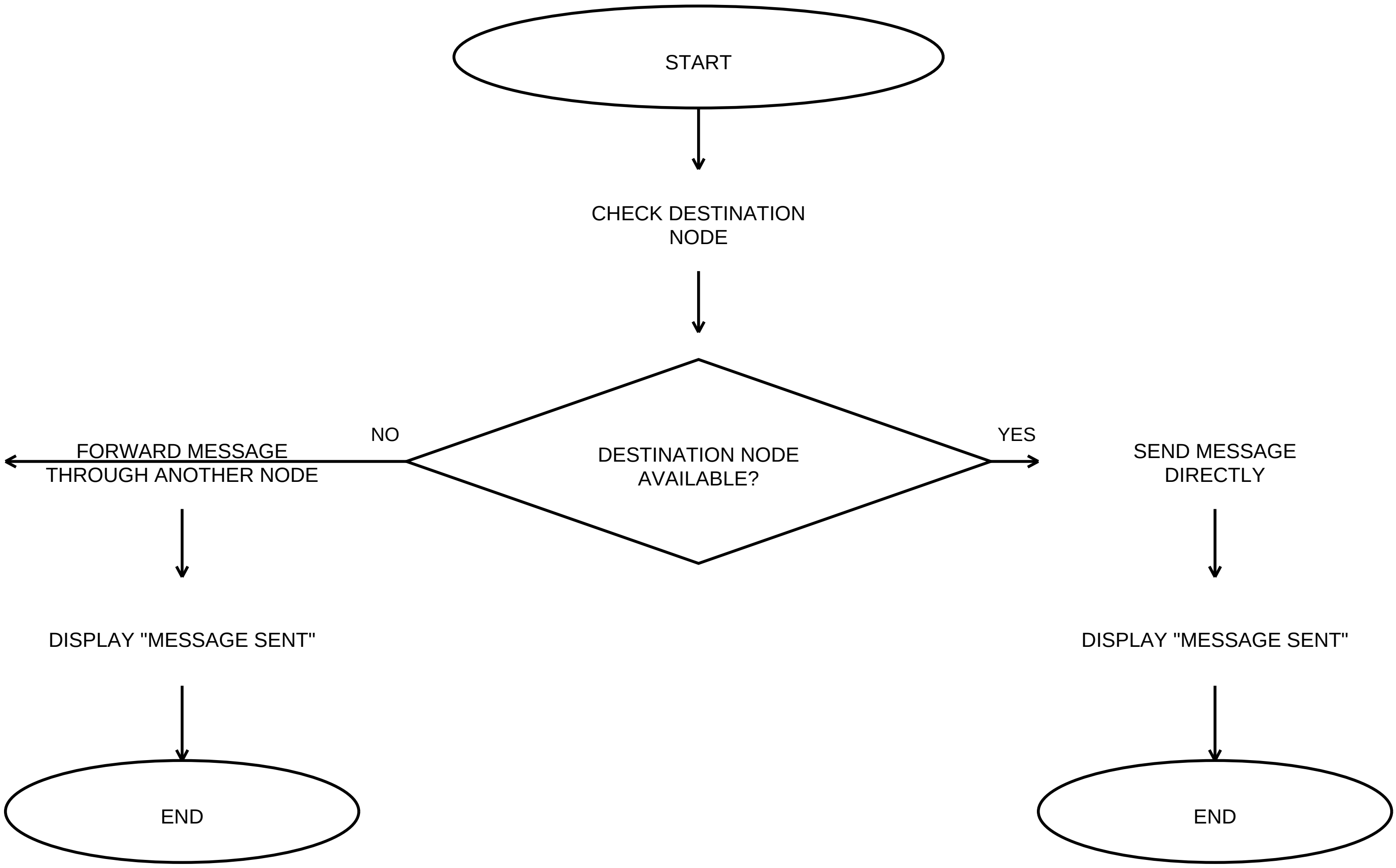
Topic: Overlay Networks

Scenario: A user wants to send a message to another node in an overlay network.

A. Correct Pseudocode Block Arrangement

1. `START`
2. `CHECK DESTINATION NODE`
3. `IF DESTINATION NODE IS AVAILABLE THEN`
4. `SEND MESSAGE DIRECTLY`
5. `ELSE`
6. `FORWARD MESSAGE THROUGH ANOTHER NODE`
7. `DISPLAY "MESSAGE SENT"`
8. `ENDIF`
9. `END`

B. Corresponding Flowchart


START
CHECK DESTINATION NODE
DESTINATION NODE AVAILABLE?
NO
YES
FORWARD MESSAGE THROUGH ANOTHER NODE
SEND MESSAGE DIRECTLY
DISPLAY "MESSAGE SENT"
DISPLAY "MESSAGE SENT"
END
END

FINAL SUBMISSION CHECKLIST

■ Fill in your Name, Register No., Section/Batch and Date.

■ Include Q1, Q2 and Q3 pseudocode in the correct sequence.

■ Include one flowchart for each question.

■ Use oval for Start/End, rectangle for Process/Action, diamond for Decision, parallelogram for Input/Output, and arrows for flow direction.

■ Check that the YES/NO branches are clearly labelled.

■ Save/export the completed work as a PDF.

■ Put the required PDF/work in your GitHub folder/repository.

■ Copy the GitHub repository/folder link and submit it in Google Classroom (GCR).

How to make the PDF: Open the content in Word or Google Docs, fill in your details, then choose Download/Save as PDF. You can also use this generated PDF as your base and fill in your details before submission.